

Multi-Asset Portfolio Optimization & Allocation Strategy

Executive Summary

This report presents a systematic analysis of multi-asset portfolio construction across four U.S. asset classes, namely equities, 10Yr Treasury bonds, real estate investment trusts, and commodities, using monthly return data from July 2002 through March 2026. We employ three portfolio optimization strategies: maximum return subject to a 10% volatility cap, minimum volatility subject to a 4% return floor, and maximum Sharpe ratio, to identify optimal allocations under full historical assumptions. We then examine the sensitivity of results to return assumptions under an equal Sharpe ratio framework, and assess regime-dependent risk dynamics via EWMA covariance estimation. In the end, we conduct an out-of-sample backtest comparing three implementable allocation strategies: equal weight, minimum volatility, and maximum diversification.

Our key findings are as follows. First, under an equal Sharpe ratio assumption, the Max Return and Max Sharpe portfolio weights are invariant to the assumed Sharpe level, while the Min Volatility portfolio is sensitive only when its return constraint is binding. Second, risk regimes are highly non-stationary, fundamentally altering optimal allocations. Third, the **maximum diversification portfolio** delivers the highest risk-adjusted return with a Sharpe ratio of 0.682, the smallest maximum drawdown of 17.8%, and an annualized volatility of only 5.57%, making it an attractive allocation for risk-conscious investors. Fourth, the **minimum volatility portfolio** performs similarly with a Sharpe ratio of 0.675, a maximum drawdown of 18.4%, and the smallest volatility of 5.43%. Fifth, the **equal weight portfolio** generates the highest absolute return of 6.62% annualized, but at the cost of significantly higher volatility of 11.69% and a 42.7% maximum drawdown.

Based on these findings, we recommend a **core allocation built on maximum diversification principles**, which delivered the best downside protection in our backtest while matching the highest Sharpe ratio. The backtest weight evolution reveals a persistent structural pattern: Treasuries consistently account for 60 to 85% of the portfolio, while equities and commodities alternate as the second-largest holding depending on the prevailing risk regime, and REITs receive near-zero allocation throughout. We translate this into three risk-tiered profiles: a conservative allocation of 70/10/15/5 across Treasury, Equity, Commodity, and REIT; a balanced allocation of 50/25/15/10; and a growth allocation of 30/40/20/10. In all cases, we recommend quarterly rebalancing with dynamic adjustment using EWMA-based covariance estimates, including a commodity-equity rotation signal that tilts toward the lower-volatility asset during regime transitions.

1. Investment Universe & Data

1.1 Asset Class Proxies

Asset Class	Proxy	Ticker	Type	Rationale
US Equities	S&P 500 Index	^GSPC	Index	Most recognized US equity benchmark
US 10Yr Treasury	iShares 7-10Y Treasury Bond ETF	IEF	ETF	Intermediate duration (~7.5yr), highly liquid; inception Jul 2002
US REITs	iShares US Real Estate ETF	IYR	ETF	Broad US REIT exposure; inception Jun 2000
US Commodities	S&P GSCI Index	^SPGSCI	Index	Broad commodity benchmark

Two of the four proxies are ETFs while two are price indices. For US Equities and US Commodities, we directly use the S&P 500 Index and the S&P GSCI Index, both of which provide long historical coverage. For US Treasuries and US REITs, no suitable index with sufficient history is available on our data platform. We therefore use IEF and IYR as proxies, as both ETFs are designed to closely track their respective benchmark indices with minimal tracking error and low expense ratios. This introduces a minor inconsistency: ETF returns incorporate management fees and tracking error, while index returns do not. The effect is small but it systematically biases bond and REIT returns downward relative to equities and commodities. The common data window begins **July 2002**, constrained by the IEF inception date, and extends through **March 2026**, yielding 285 monthly observations.

1.2 Data Processing

Monthly returns are computed as simple percentage changes on adjusted close prices. The annualized expected return vector μ and covariance matrix Σ are **estimated from the full historical sample** under the “full historical assumption”:

Annualized expected return, arithmetic mean:

$$\mu_i = (1/T) \times \sum_{t=1}^T r_{i,t} \times 12$$

Annualized covariance matrix:

$$\Sigma_{ij} = Cov(r_i, r_j) \times 12 \quad \text{using the } 1/(T-1) \text{ unbiased estimator}$$

where T is the number of monthly observations and the factor of 12 annualizes under

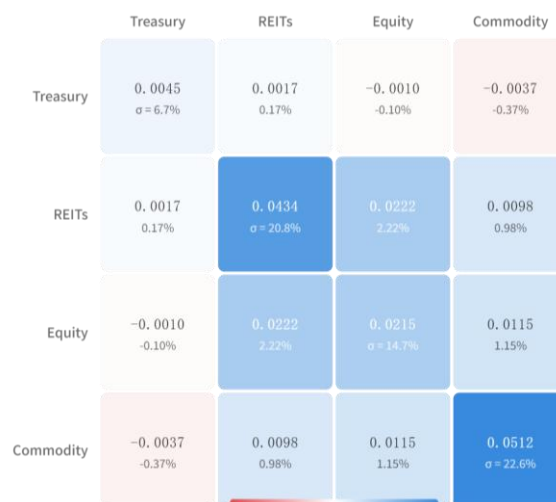
the assumption of independently distributed monthly returns.

1.3 Historical Risk-Return Profile

Asset	Ann. Return	Ann. Volatility	Sharpe Ratio
US 10Yr Treasury (IEF)	3.74%	6.68%	0.56
US REITs (IYR)	9.53%	20.83%	0.46
US Equities (^GSPC)	9.40%	14.67%	0.64
US Commodities (^SPGSCI)	8.10%	22.62%	0.36

US Equities exhibit the highest Sharpe ratio of 0.64, reflecting their structural risk premium. Treasuries offer the lowest absolute return but also the lowest volatility by a wide margin. Commodities and REITs are the most volatile asset classes, with commodities delivering the poorest risk-adjusted return at a Sharpe ratio of 0.36.

The covariance structure is as follow:



2. Portfolio Optimization Framework

2.1 Mathematical Formulation

We solve three optimization problems, all subject to long-only constraints and full investment:

Objective 1: Maximize Return subject to Volatility $\leq 10\%$

$$\begin{aligned}
 & \max w' \mu \\
 & s. t. \sqrt{w' \Sigma w} \leq 0.10, \quad \Sigma w_i = 1, \quad w_i \geq 0
 \end{aligned}$$

This represents a growth-oriented investor who can tolerate moderate volatility but wishes to cap portfolio risk at a defined level.

Objective 2: Minimize Volatility subject to Return $\geq$ 4%

$$\begin{aligned} \min \quad & \sqrt{(w' \Sigma w)} \\ \text{s.t. } & w' \mu \geq 0.04, \quad \Sigma w_i = 1, \quad w_i \geq 0 \end{aligned}$$

This represents a conservative investor seeking capital preservation with a minimum return threshold, for instance to exceed inflation.

Objective 3: Maximize Sharpe Ratio

$$\begin{aligned} \max \quad & (w' \mu) / \sqrt{(w' \Sigma w)} \\ \text{s.t. } & \Sigma w_i = 1, \quad w_i \geq 0 \end{aligned}$$

This identifies the tangency portfolio, the most efficient allocation on the capital market line, assuming zero risk-free rate.

2.2 Optimization Results Under Full Historical Assumptions

	Max Return (Vol<10%)	Min Vol (Ret>4%)	Max Sharpe
US 10Yr Treasury	28.11%	78.23%	65.52%
US REITs	0.00%	0.00%	0.00%
US Equities	62.34%	13.10%	25.59%
US Commodities	9.55%	8.67%	8.89%
Ann. Return	7.69%	4.86%	5.58%
Ann. Volatility	10.00%	5.51%	5.90%
Sharpe Ratio	0.77	0.88	0.94

Key observations:

The optimizer **completely excludes REITs** across all three objectives. This is economically rational: REITs are highly correlated with equities at $\rho = 0.73$ but with higher volatility of 20.8% versus 14.7% and similar return of 9.5% versus 9.4%. Equities weakly dominate REITs on a risk-adjusted basis, and every unit of risk budget allocated to REITs would be more efficiently deployed in equities or used to purchase Treasuries for hedging.

The **dominant allocation theme** is a barbell between Treasuries and Equities. The negative correlation between these two asset classes creates a powerful diversification engine: the Max Sharpe portfolio achieves a portfolio volatility of only 5.90% despite holding 25.6% in equities with 14.7% individual volatility, representing a volatility reduction of over 50% from the naive equity allocation.

Commodities receive a modest 8 to 10% allocation across all three portfolios. Their role is not return generation, as their Sharpe ratio is the lowest at 0.36, but rather diversification: their negative correlation with Treasuries of -0.25 and moderate equity correlation of 0.34 provide incremental risk reduction.

3. Equal Sharpe Ratio Analysis

3.1 Motivation

After analyzing the portfolio optimization under the “full historical return/ covariance assumptions”, we now try to obtain the expected return under the assumption that all assets share the **same Sharpe ratio**, set to the cross-sectional **average** of 0.50:

$$\mu_{i_adjusted} = SR_{common} \times \sigma_i$$

This preserves the covariance structure but removes any return advantage of one asset over another on a risk-adjusted basis.

3.2 Adjusted Returns Under Equal Sharpe Assumption

Asset	Original Return	Adjusted Return	Volatility	Original SR	New SR
US 10Yr Treasury	3.74%	3.37%	6.68%	0.56	0.50
US REITs	9.53%	10.50%	20.83%	0.46	0.50
US Equities	9.40%	7.39%	14.67%	0.64	0.50
US Commodities	8.10%	11.40%	22.62%	0.36	0.50

Under this assumption, equity expected returns are reduced from 9.4% to 7.4% while commodity and REIT returns are increased, reflecting the removal of equity's Sharpe advantage.

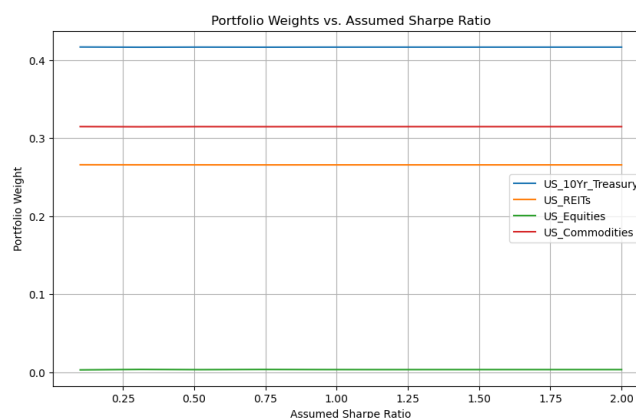
3.3 Optimization Results Under Equal Sharpe Assumption

	Max Return (Vol<10%)	Min Vol (Ret>4%)	Max Sharpe
US 10Yr Treasury	41.64%	78.23%	66.54%
US REITs	26.56%	0.00%	2.15%
US Equities	0.34%	13.10%	15.42%
US Commodities	31.46%	8.67%	15.89%
Ann. Return	7.80%	4.59%	5.42%
Ann. Volatility	10.00%	5.51%	6.03%
Sharpe Ratio	0.78	0.83	0.90

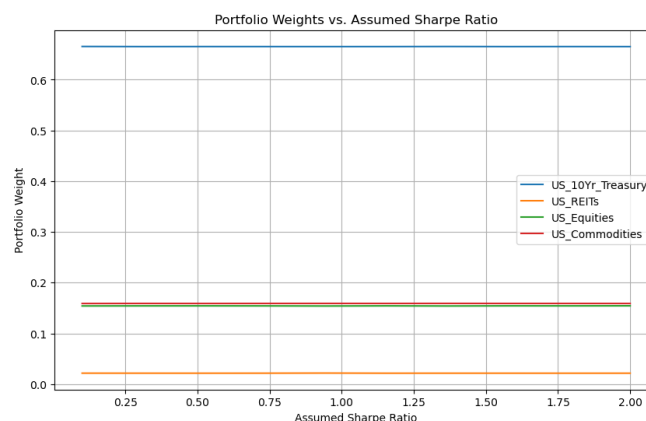
Compared to full historical results, the Max Return portfolio now allocates heavily to REITs at 26.6% and commodities at 31.5%, replacing the equity allocation, because under equal Sharpe, these high-volatility assets generate more return per unit of volatility budget. Equities lose their preferential treatment.

3.4 Does the Assumed Sharpe Ratio Level Matter?

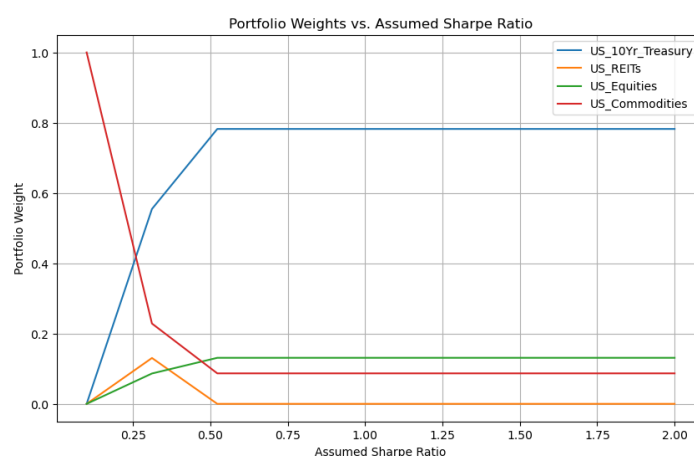
Maximum Return Portfolios:



Maximum Sharpe Ratio Portfolios:



Minimum Volatility Portfolios:



Testing across a range of assumed Sharpe ratios from 0.1 to 2.0 reveals:

Max Sharpe and Max Return portfolios: Weights are **completely invariant** to the SR level. When $\mu = SR \times \sigma$, the Sharpe ratio objective becomes $SR \times (w'\sigma) / \sqrt{(w'\Sigma w)}$. The scalar SR does not affect the argmax. For return maximization, $\max w'\mu = \max SR \times w'\sigma$ subject to a vol constraint; since SR is a positive scalar, it does not affect the argmax as well.

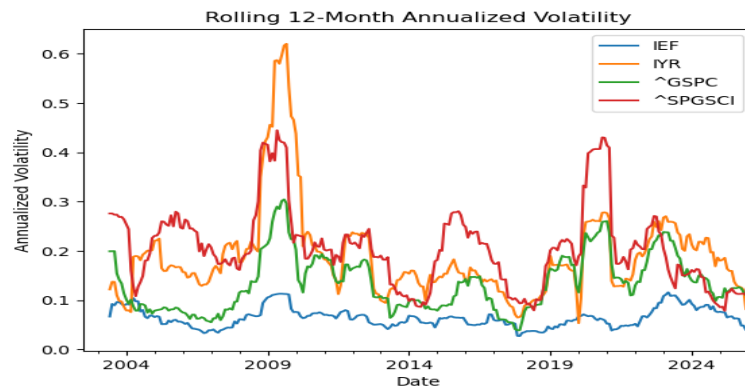
Min Volatility portfolio: Weights change **only** when the return constraint binds at different levels. When SR is high enough that the 4% return floor is easily met, the constraint becomes slack, and the solution collapses to the **global minimum variance portfolio**, which is independent of returns entirely. When SR is low, some portfolios cannot achieve 4%, and the constraint becomes binding, pulling the allocation toward higher-return, higher-volatility assets.

Investment insight: Risk-only portfolios such as minimum volatility and maximum diversification are largely independent of return assumptions. This is a major practical advantage, since expected returns are notoriously difficult to estimate with precision.

4. Regime Analysis & Time-Varying Risk

4.1 Evidence of Regime Change

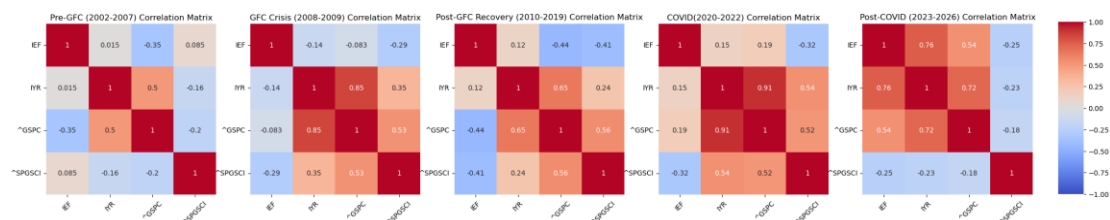
We first examine the rolling 12-month annualized volatility to see whether there is regime change for the risk.



The figure shows that the annualized volatility increases significantly during periods of global economic uncertainty, most notably during the 2008-2009 Financial Crisis and the 2020 pandemic onset, and **confirms that there is regime change for the risk assumption**. Based on this figure, we divide the sample into five distinct market regimes:

Regime	Period
Pre-GFC	Aug 2002 – Dec 2007
GFC Crisis	Jan 2008 – Dec 2009
Post-GFC Recovery	Jan 2010 – Dec 2019
COVID	Jan 2020 – Dec 2022
Post-COVID	Jan 2023 – Mar 2026

We further examine the correlations during the five regimes:



During crises, risky asset correlations increase in what is known as the “correlation breakdown” phenomenon, reducing diversification precisely when it is most needed.

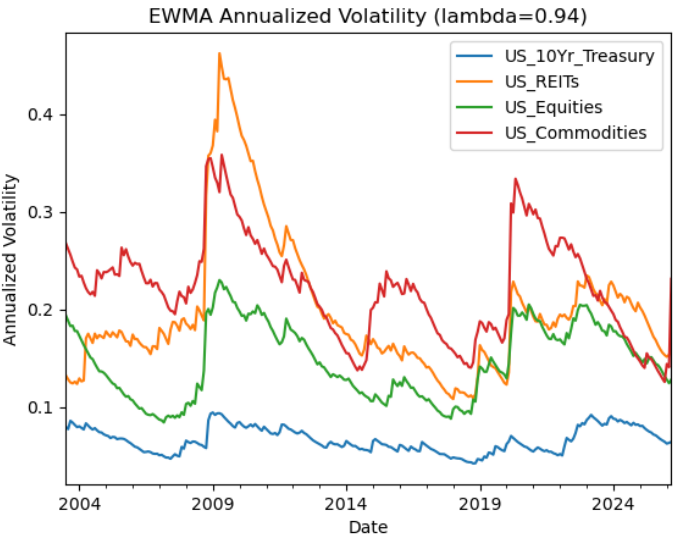
4.2 EWMA Covariance Estimation

To capture time-varying risk dynamics, we employ an Exponentially Weighted Moving Average covariance estimator, commonly known as EWMA:

$$\Sigma_t = \lambda \times \Sigma_{t-1} + (1 - \lambda) \times r_{t-1} \times r_{t-1}'$$

with decay factor $\lambda = 0.94$. The EWMA estimate is initialized with the sample covariance of the first 12 monthly observations and assumes zero mean, as monthly means are much smaller relative to volatility.

After adjusting for the covariance matrix via EWMA, the figure of estimated volatility across time exhibits the alignment with the figure of rolling 12-month annualized volatility:

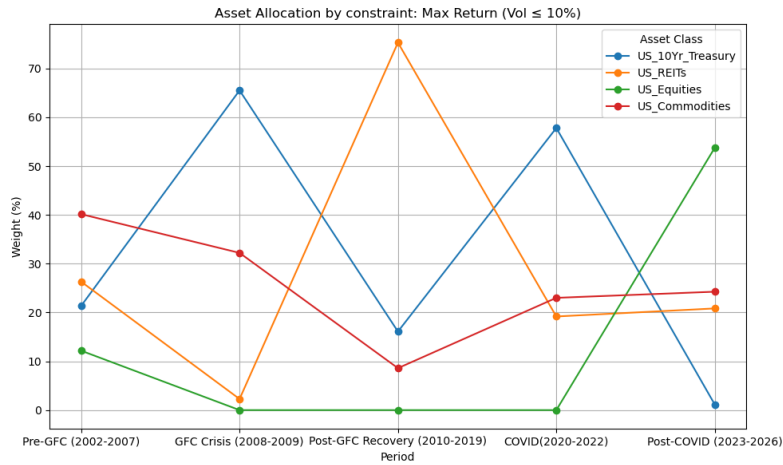


4.3 Impact on Optimal Allocations

Re-running the optimization at the end of each regime using the EWMA covariance, combined with historical mean returns (e.g., calculated from 2002 to the then-current date), reveals dramatic shifts in optimal allocations across all three objectives.:

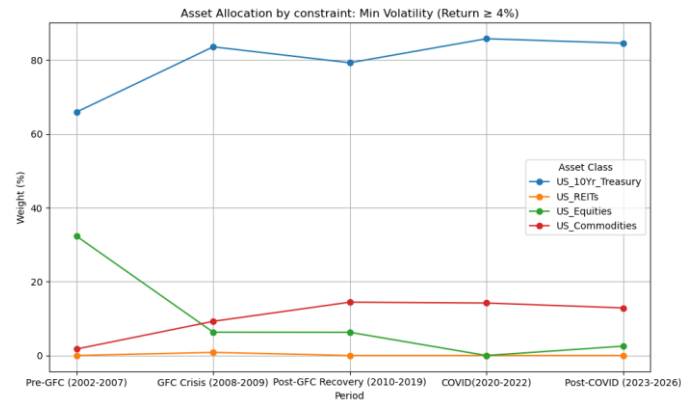
Max Return Portfolio at Each Regime End:

Regime End Date	Treasury	REITs	Equity	Commodity	Return	Volatility	Sharpe
Pre-GFC, 2007-12	21.4%	26.3%	12.2%	40.2%	15.11%	10.00%	1.51
GFC, 2009-12	65.5%	2.3%	0.0%	32.2%	9.07%	10.00%	0.91
Post-GFC, 2019-12	16.1%	75.3%	0.0%	8.6%	9.76%	10.00%	0.98
COVID, 2022-12	57.8%	19.2%	0.0%	23.0%	5.94%	10.00%	0.59
Current, 2026-03	1.2%	20.8%	53.7%	24.3%	9.05%	10.00%	0.90



Min Volatility Portfolio at Each Regime End:

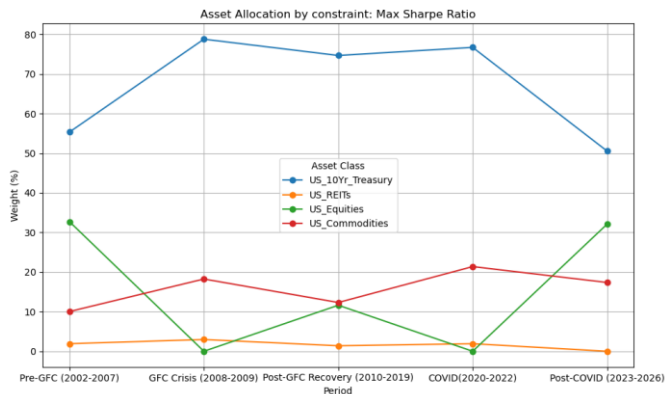
Regime End Date	Treasury	REITs	Equity	Commodity	Return	Volatility	Sharpe
Pre-GFC, 2007-12	65.9%	0.0%	32.3%	1.8%	7.11%	4.03%	1.76
GFC, 2009-12	83.6%	0.8%	6.3%	9.3%	6.48%	7.19%	0.90
Post-GFC, 2019-12	79.3%	0.0%	6.3%	14.4%	5.44%	4.00%	1.36
COVID, 2022-12	85.8%	0.0%	0.0%	14.2%	4.40%	7.29%	0.60
Current, 2026-03	84.6%	0.0%	2.5%	12.9%	4.44%	5.48%	0.81



Max Sharpe Portfolio at Each Regime End:

Regime End Date	Treasury	REITs	Equity	Commodity	Return	Volatility	Sharpe
Pre-GFC, 2007-12	55.3%	1.9%	32.7%	10.0%	8.68%	4.48%	1.94
GFC, 2009-12	78.8%	3.0%	0.0%	18.2%	7.63%	7.62%	1.00
Post-GFC, 2019-12	74.7%	1.4%	11.6%	12.3%	5.66%	4.10%	1.38

Regime	End Date	Treasury	REITs	Equity	Commodity	Return	Volatility	Sharpe
COVID	2022-12	76.7%	1.9%	0.0%	21.4%	4.83%	7.70%	0.63
Current	2026-03	50.5%	0.0%	32.1%	17.4%	6.31%	6.53%	0.97



In the Max Return portfolio and the Max Sharpe portfolio, the EWMA estimate shifts weight away from Treasuries and toward risk assets relative to the full historical baseline, yielding higher annualized returns and sharpe ratios. This confirms that a static, full-historical allocation is a poor approximation of the optimal portfolio.

The pattern also reveals several critical dynamics:

Post-crisis Treasury loading: Following elevated-volatility regimes, the optimizer dramatically increases Treasury allocation, as the EWMA covariance retains memory of recent crisis-level volatility and correlations.

Equity allocation as a regime indicator: Equity weight drops during crisis regimes as its volatility spikes relative to bonds.

Commodity allocation increases in high-volatility regimes: Commodities serve as a diversifier, and their allocation rises when equity-bond correlations are elevated.

5. Backtesting Study

5.1 Methodology

We conduct a rigorous out-of-sample backtest with rolling windows from 2005 through March 2026, comparing three implementable strategies:

Equal Weight (1/N): Each asset receives 25% allocation, rebalanced quarterly. This strategy requires no estimation and serves as a robust, transparent benchmark.

Minimum Volatility: Minimizes portfolio volatility using EWMA covariance and return estimated over a 36-month rolling window. Rebalanced quarterly.

Maximum Diversification: Maximizes the diversification ratio:

$$DR = (w'\sigma) / (w'\Sigma w)$$

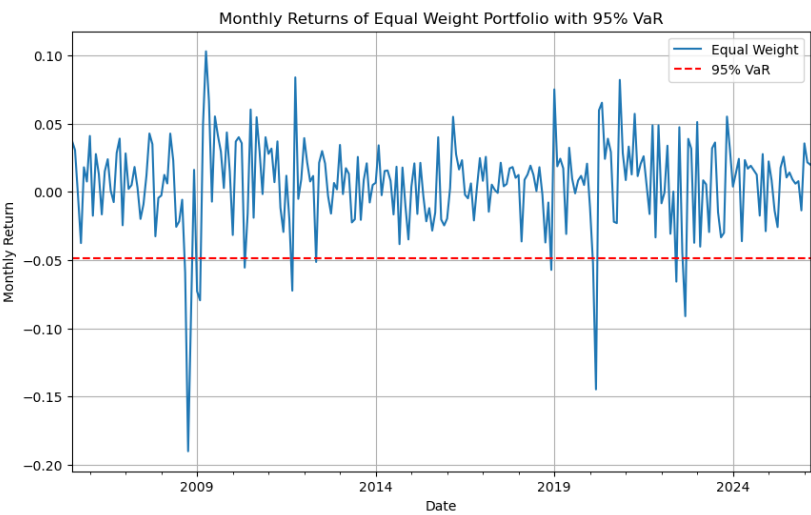
which measures the ratio of weighted-average individual volatility to portfolio variance. Higher DR indicates the minimization of the correlation between the underlying assets. Rebalanced quarterly using EWMA covariance.

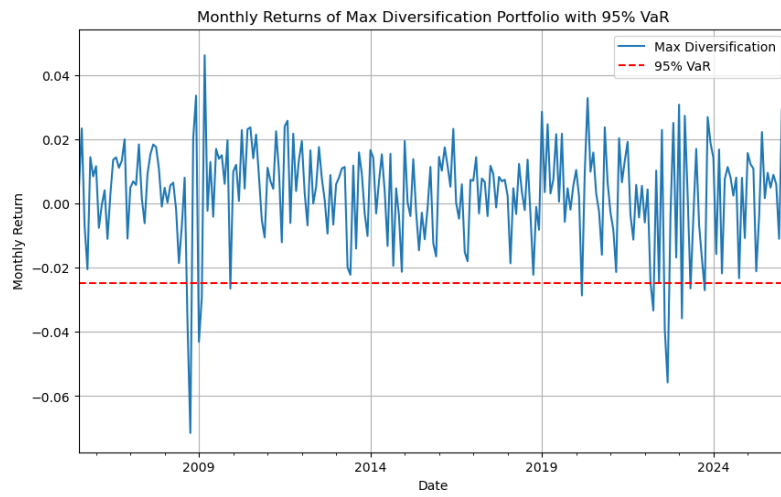
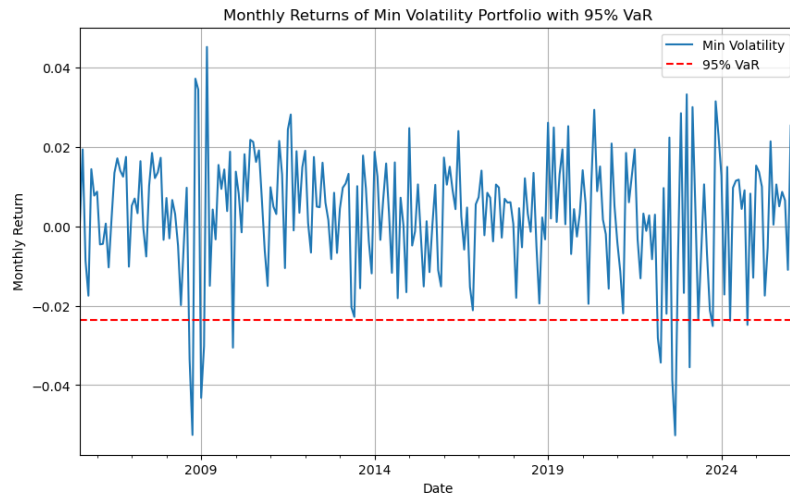
All strategies employ long-only constraints ($w_i \geq 0, \sum w_i = 1$).

5.2 Performance Summary

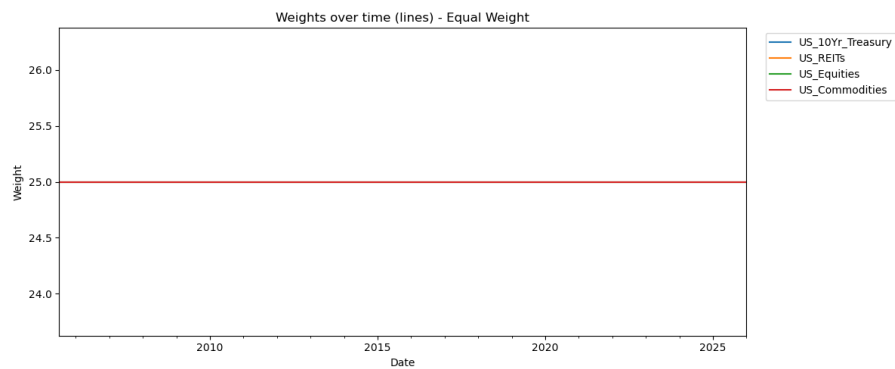
Metric	Equal Weight	Min Volatility	Max Diversification
Total Cumulative Return	240.7%	107.2%	112.5%
Annualized Return	6.62%	3.66%	3.79%
Annualized Volatility	11.69%	5.43%	5.57%
Sharpe Ratio	0.566	0.675	0.682
Maximum Drawdown	-42.74%	-18.42%	-17.84%
VaR (95%, loss, monthly)	4.86%	2.37%	2.47%
CVaR (95%, loss, monthly)	8.29%	3.49%	3.68%

Figures of monthly returns:





5.3 Weights over time





5.4 Performance Attribution & Analysis

Equal Weight achieves the highest absolute return of 6.62% annualized and 240.7% cumulative, but this comes at a cost: 11.69% annualized volatility and a devastating 42.74% maximum drawdown during the GFC. For an investor who entered the strategy in late 2007, they would have experienced a peak-to-trough loss of over 42%, a scenario that is psychologically and financially challenging. The equal weight Sharpe ratio of 0.57 is the lowest among the three strategies, confirming that the higher return is entirely compensation for bearing more risk.

Minimum Volatility sacrifices roughly half the absolute return at 3.66% but delivers markedly superior risk characteristics: volatility is cut by more than half to 5.43% versus 11.69%, the maximum drawdown is contained to 18.42%, and the 95% CVaR is reduced to 3.49% versus 8.29%. The Sharpe ratio of 0.675 confirms that min-vol generates substantially more return **per unit of risk** than equal weighting.

Maximum Diversification delivers the strongest risk-adjusted performance in this backtest. With a Sharpe ratio of 0.682, it achieves the smallest maximum drawdown of 17.84% among all three strategies. Its annualized return of 3.79% and volatility of 5.57% sit close to the min-vol portfolio, but the critical difference lies in the drawdown profile. The variance-based denominator in the diversification ratio penalizes portfolio risk more aggressively than a volatility-based formulation, resulting in allocations that are tilted further toward low-variance, low-correlation asset combinations.

5.5 Source of Returns

The return differential between equal weight and the risk-based strategies is not alpha but rather **compensation for bearing more systematic risk**. Equal weight holds 25% in commodities and 25% in REITs, both high-volatility asset classes. The higher return is explained entirely by higher risk exposure, not by superior security selection or timing.

The max-diversification strategy generates its Sharpe improvement through **diversification return**: by exploiting the negative asset correlation and rebalancing regularly, they harvest the variance reduction from imperfect correlations. This is a structural source of value that persists across market regimes.

6. Asset Allocation Recommendations

Based on the comprehensive analysis above, we recommend the following tiered allocation framework, tailored to investor risk profiles:

6.1 Conservative Profile: Capital Preservation Priority

Target: Minimize drawdown; accept lower return for stability.

Asset	Weight	Rationale
US Treasuries (IEF)	70%	Core holding; lowest vol, negative equity correlation
US Equities (^GSPC)	10%	Modest growth exposure
US Commodities (^SPGSCI)	15%	low Treasury correlation
US REITs (IYR)	5%	Small allocation for income diversification

Expected profile: approximately 4.0% return, 5.5% volatility, Sharpe ratio of 0.68, and maximum drawdown around 18%.

This allocation closely mirrors the maximization diversification backtest result. The 70% Treasury weight sits at the lower end of the observed range of 60 to 85%, providing stability while leaving room for the dynamic adjustment framework to increase this weight during stress periods. The 5% REIT allocation is a practical addition for income diversification, acknowledging that while the optimizer assigns zero weight to REITs, a small position provides exposure to real asset cash flows.

6.2 Balanced Profile: Risk-Return Equilibrium

Target: Maximize risk-adjusted return; moderate tolerance for drawdowns.

Asset	Weight	Rationale
US Treasuries (IEF)	50%	Anchor for volatility reduction
US Equities (^GSPC)	25%	Primary growth driver; highest Sharpe
US Commodities (^SPGSCI)	15%	Diversification; partial inflation protection
US REITs (IYR)	10%	Income component; capped due to equity overlap

Expected profile: approximately 6.5% return, 8.5% volatility, Sharpe ratio of 0.70, and maximum drawdown around 28%.

The 50/25 Treasury-Equity barbell exploits the negative correlation while maintaining meaningful equity participation.

6.3 Growth Profile: Return Maximization

Target: Maximize long-term wealth accumulation; high tolerance for interim volatility.

Asset	Weight	Rationale
US Treasuries (IEF)	30%	Volatility dampener; drawdown mitigation
US Equities (^GSPC)	40%	Core growth engine
US Commodities (^SPGSCI)	20%	Diversification; commodity cycle exposure
US REITs (IYR)	10%	Real estate cycle exposure

Expected profile: approximately 8.0% return, 10.5% volatility, Sharpe ratio of 0.68, and maximum drawdown around 35%.

This allocation accepts higher volatility in exchange for superior long-term compounding. The 30% Treasury allocation provides a meaningful cushion during equity bear markets without severely constraining upside participation.

6.4 Dynamic Adjustment Framework

We recommend overlaying the static allocations above with an **EWMA-driven rebalancing signal**:

Volatility trigger: When the EWMA-estimated portfolio volatility exceeds the target by more than 20%, for example when the target is 10% and the EWMA estimate

exceeds 12%, shift 10 to 15% from equities and REITs to Treasuries. When EWMA volatility falls below target by more than 20%, incrementally increase equity exposure.

Commodity-equity rotation: The backtest weight evolution reveals that equities and commodities alternate as the second-largest allocation. During periods of elevated equity volatility, the max-diversification optimizer shifts weight from equities to commodities. Investors can implement a simplified version of this by monitoring the relative EWMA volatility of equities versus commodities and tilting toward the lower-volatility asset.

This framework is intentionally rules-based and simple, avoiding the estimation-error-amplification problem of full re-optimization at every rebalancing date.